

## FLODA, Sweden

WMO: 024760

Lat: 59.0558N

Lon: 16.3944E

Elev: 108

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.9       | 5.0 | -5.7                          | 4.1 | -1.4 | 1.4 | 5.9 | 5.8  | 24.7                  | 42.9 | 21.5 | 40.0 | 1.3                   | 300  | 0.586 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.7                   | 80.7            | 65.4 | 77.5 | 64.0 | 74.4 | 62.3 | 67.7                | 76.4 | 65.8 | 74.0 | 64.0 | 71.8 | 6.8                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.8                            | 92.2 | 70.7 | 62.8 | 86.0 | 69.1 | 60.8 | 79.8 | 67.0 | 32.1          | 76.5 | 30.6 | 74.0 | 29.3 | 71.7 | 73.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| 19.6              | 16.7 | 14.4 | DB                         | -8.1 | 84.3               | 8.4 | 3.5                                                | -14.1 | 86.9       | -19.0 | 89.0       | -23.7 | 90.9       | -29.8 | 93.5 | (4) |
|                   |      |      | WB                         | -8.6 | 70.3               | 8.2 | 2.2                                                | -14.5 | 72.0       | -19.3 | 73.3       | -23.9 | 74.5       | -29.9 | 76.2 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 43.9  | 27.1  | 28.0  | 33.1  | 41.9  | 51.0  | 57.9  | 62.6  | 60.7  | 53.0  | 43.9  | 36.3  | 30.4 | (6)  |
|      |                                                                   | DBStd       | 14.39 | 10.00 | 9.34  | 8.19  | 6.34  | 6.27  | 5.06  | 4.65  | 4.83  | 5.72  | 6.47  | 7.18  | 9.73 | (7)  |
|      |                                                                   | HDD50       | 3435  | 711   | 617   | 524   | 251   | 64    | 0     | 1     | 36    | 207   | 411   | 609   |      | (8)  |
|      |                                                                   | HDD65       | 7752  | 1176  | 1037  | 989   | 692   | 436   | 220   | 105   | 148   | 360   | 655   | 860   | 1074 | (9)  |
|      |                                                                   | CDD50       | 1215  | 0     | 0     | 0     | 10    | 94    | 240   | 392   | 334   | 127   | 17    | 1     | 0    | (10) |
|      |                                                                   | CDD65       | 54    | 0     | 0     | 0     | 0     | 6     | 32    | 16    | 0     | 0     | 0     | 0     | 0    | (11) |
|      |                                                                   | CDH74       | 745   | 0     | 0     | 0     | 0     | 28    | 117   | 417   | 180   | 3     | 0     | 0     | 0    | (12) |
|      |                                                                   | CDH80       | 118   | 0     | 0     | 0     | 0     | 1     | 13    | 86    | 18    | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 6.3   | 6.4   | 6.5   | 6.9   | 6.5   | 6.6   | 6.2   | 5.7   | 5.6   | 6.1   | 6.3   | 6.3   | 6.6  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 23.5  | 1.5   | 1.3   | 1.1   | 1.4   | 1.7   | 2.4   | 2.9   | 2.9   | 2.3   | 2.0   | 2.1   | 1.8  | (15) |
| (16) |                                                                   | PrecMax     | 29.0  | 3.0   | 2.8   | 2.3   | 3.9   | 3.3   | 4.9   | 6.7   | 7.4   | 6.1   | 4.7   | 4.8   | 3.7  | (16) |
| (17) |                                                                   | PrecMin     | 17.0  | 0.4   | 0.5   | 0.2   | 0.2   | 0.5   | 0.4   | 0.6   | 0.1   | 0.5   | 0.0   | 0.6   | 0.5  | (17) |
| (18) |                                                                   | PrecStd     | 2.4   | 0.7   | 0.6   | 0.5   | 0.9   | 0.8   | 1.1   | 1.3   | 1.5   | 1.5   | 1.0   | 1.0   | 0.8  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 46.4  | 47.5  | 57.7  | 68.8  | 77.6  | 81.7  | 86.6  | 82.3  | 74.0  | 61.5  | 54.0  | 50.9  |      | (19) |
| (20) |                                                                   | 0.4% MCWB   | 44.3  | 43.6  | 47.6  | 54.0  | 59.6  | 64.5  | 66.8  | 65.7  | 61.9  | 56.1  | 51.1  | 48.1  |      | (20) |
| (21) |                                                                   | 2% DB       | 43.3  | 44.3  | 51.9  | 63.4  | 73.2  | 77.3  | 82.3  | 78.8  | 69.8  | 58.2  | 50.9  | 46.8  |      | (21) |
| (22) |                                                                   | 2% MCWB     | 41.1  | 41.0  | 43.9  | 51.0  | 58.0  | 62.9  | 66.0  | 65.2  | 60.4  | 54.3  | 48.8  | 44.7  |      | (22) |
| (23) |                                                                   | 5% DB       | 40.8  | 41.7  | 47.8  | 58.9  | 69.1  | 73.9  | 78.7  | 75.4  | 66.4  | 55.8  | 48.5  | 44.2  |      | (23) |
| (24) |                                                                   | 5% MCWB     | 38.9  | 38.9  | 41.7  | 48.9  | 56.2  | 60.7  | 64.6  | 63.6  | 58.7  | 52.6  | 46.6  | 42.3  |      | (24) |
| (25) |                                                                   | 10% DB      | 38.7  | 39.4  | 44.2  | 54.5  | 64.7  | 70.2  | 75.2  | 72.0  | 63.2  | 53.7  | 46.2  | 41.9  |      | (25) |
| (26) |                                                                   | 10% MCWB    | 37.1  | 37.1  | 39.6  | 45.8  | 53.7  | 58.9  | 63.8  | 61.8  | 57.0  | 51.0  | 44.7  | 40.2  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 44.7  | 44.4  | 48.1  | 56.0  | 62.9  | 67.5  | 70.3  | 69.7  | 64.6  | 57.7  | 51.8  | 48.3  |      | (27) |
| (28) |                                                                   | 0.4% MCDB   | 46.5  | 47.0  | 56.2  | 65.7  | 72.4  | 76.2  | 79.0  | 77.3  | 70.7  | 60.2  | 53.3  | 50.2  |      | (28) |
| (29) |                                                                   | 2% WB       | 41.6  | 41.5  | 44.7  | 52.0  | 60.1  | 64.5  | 68.2  | 67.2  | 61.5  | 55.0  | 49.0  | 45.0  |      | (29) |
| (30) |                                                                   | 2% MCDB     | 43.0  | 43.8  | 50.6  | 61.0  | 68.9  | 73.8  | 77.5  | 74.5  | 67.7  | 57.1  | 50.6  | 46.3  |      | (30) |
| (31) |                                                                   | 5% WB       | 39.1  | 39.3  | 42.4  | 49.6  | 57.3  | 62.1  | 66.5  | 65.2  | 59.6  | 53.2  | 46.8  | 42.5  |      | (31) |
| (32) |                                                                   | 5% MCDB     | 40.6  | 41.3  | 47.0  | 57.5  | 66.5  | 71.1  | 75.7  | 72.0  | 64.8  | 55.4  | 48.3  | 43.9  |      | (32) |
| (33) |                                                                   | 10% WB      | 37.3  | 37.4  | 40.1  | 47.0  | 55.1  | 60.0  | 64.7  | 63.4  | 57.9  | 51.3  | 44.9  | 40.2  |      | (33) |
| (34) |                                                                   | 10% MCDB    | 38.6  | 39.3  | 43.7  | 53.2  | 63.7  | 67.8  | 73.4  | 70.0  | 62.3  | 53.5  | 46.0  | 41.6  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 9.7   | 11.4  | 14.8  | 18.8  | 20.4  | 19.3  | 19.7  | 19.6  | 17.4  | 12.5  | 8.3   | 8.8   |      | (35) |
| (36) |                                                                   | 5% DB MCDBR | 8.6   | 10.5  | 19.8  | 27.7  | 28.8  | 26.7  | 27.3  | 25.1  | 21.9  | 14.4  | 11.0  | 8.6   |      | (36) |
| (37) |                                                                   | 5% DB MCWBR | 8.1   | 8.2   | 13.0  | 16.6  | 15.7  | 13.9  | 13.3  | 13.1  | 13.5  | 10.8  | 9.8   | 8.1   |      | (37) |
| (38) |                                                                   | 5% WB MCDBR | 8.4   | 9.5   | 18.3  | 24.5  | 23.5  | 23.4  | 23.2  | 20.9  | 19.7  | 12.8  | 10.7  | 8.5   |      | (38) |
| (39) |                                                                   | 5% WB MCWBR | 8.1   | 8.0   | 12.6  | 15.3  | 14.0  | 13.5  | 12.3  | 12.1  | 12.7  | 10.5  | 9.7   | 8.2   |      | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.279 | 0.319 | 0.338 | 0.376 | 0.371 | 0.361 | 0.387 | 0.379 | 0.345 | 0.335 | 0.306 | 0.264 |      | (40) |
| (41) |                                                                   | taud        | 2.224 | 2.287 | 2.356 | 2.317 | 2.396 | 2.447 | 2.394 | 2.414 | 2.493 | 2.424 | 2.279 | 2.156 |      | (41) |
| (42) |                                                                   | Ebn at Noon | 169   | 220   | 254   | 262   | 272   | 277   | 266   | 259   | 251   | 214   | 156   | 129   |      | (42) |
| (43) |                                                                   | Edh at Noon | 14    | 22    | 28    | 34    | 34    | 32    | 34    | 31    | 24    | 19    | 14    | 10    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 102   | 297   | 775   | 1254  | 1642  | 1826  | 1669  | 1324  | 878   | 393   | 131   | 69    |      | (44) |
| (45) |                                                                   | RadStd      | 15    | 47    | 88    | 125   | 141   | 99    | 162   | 128   | 102   | 62    | 29    | 8     |      | (45) |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
|                         |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (59 neighbors) |  | +0.67 | N/A     | N/A    | N/A     | N/A   | N/A   | -223        | -270  | N/A   | N/A   |

Nomenclature: See separate page